Deep Learning Project Report

With the increasing use of drones in various sectors, distinguishing between drones and birds has become crucial for airspace safety, wildlife monitoring, and surveillance systems. Manual monitoring is inefficient and prone to errors, hence an automated detection system is required.

This project presents a deep learning-based solution that uses **Convolutional Neural Networks (CNN)** and **YOLOv8 with Transfer Learning** to detect and classify birds and drones in images. The system is further integrated with a **Streamlit-based interactive dashboard** for real-time detection.

CNN, Transfer Learning & YOLO Models for Image Classification & Detection

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1. Introduction

This project focuses on implementing Deep Learning techniques for image-based tasks using:

- **Convolutional Neural Networks (CNN)**
- **Transfer Learning Models**
- **YOLO (You Only Look Once) for Object Detection**

The goal is to build efficient models for classification and detection while evaluating their performance using standard metrics.

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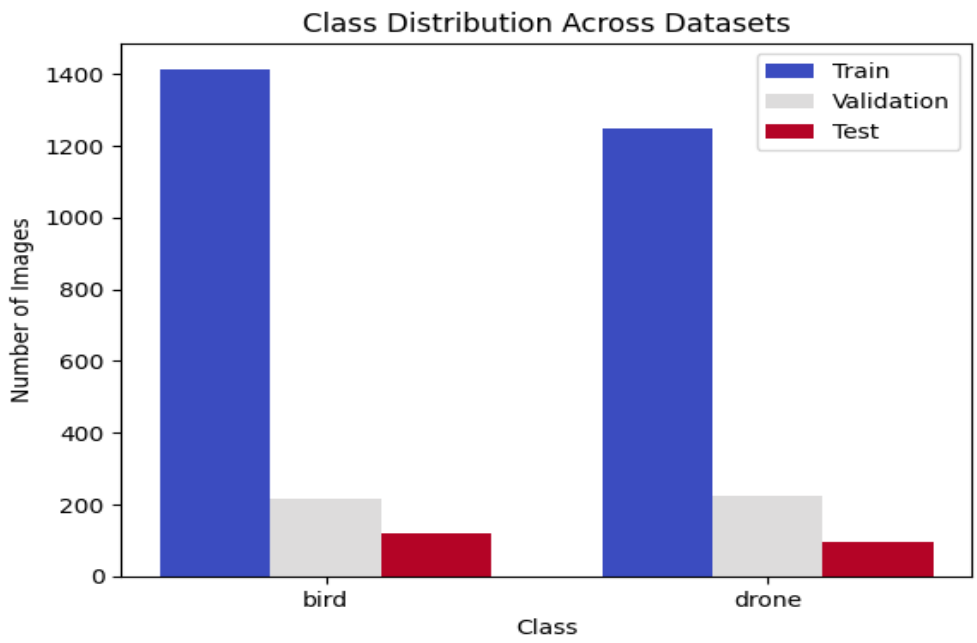
2. Dataset

The dataset consists of labeled images divided into:

- **Training Dataset** → Used for learning
- **Validation Dataset** → Used during training for tuning
- **Test Dataset** → Used for final evaluation

For YOLO:

- Images are paired with **bounding box annotations**



3. Methodology

3.1 Data Preprocessing

- Image resizing (224×224 for CNN/Transfer Learning)
- Normalization
- Dataset splitting (Train / Validation / Test)
- Label encoding (Binary classification)

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4. Model Architectures

4.1 Custom CNN Model

A basic CNN model was implemented using:

- Conv2D layers (feature extraction)
- MaxPooling layers (dimensionality reduction)
- Flatten layer
- Dense layers (classification)
- Dropout (to prevent overfitting)

4.2 Transfer Learning Models

Pre-trained models were used to improve accuracy:

Models Used:

- MobileNetV2
- ResNet50
- EfficientNetB0

Approach:

- Loaded pre-trained weights (ImageNet)
- Froze base layers (`trainable = False`)
- Added custom classification layers:
  - GlobalAveragePooling
  - Dense layer
  - Dropout
  - Output layer (sigmoid)

4.3 YOLO Model (Object Detection)

YOLO (You Only Look Once) is used for real-time object detection.

Key Features:

- Detects objects in a single forward pass
- Predicts:
  - Bounding boxes
  - Class labels

Workflow:

- Dataset preparation (images + labels)
- Training using YOLO framework
- Evaluation using:
  - mAP (mean Average Precision)
  - Precision & Recall

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5. Model Compilation

All classification models were compiled using:

- **Optimizer**: Adam
- **Learning Rate**: 0.0001
- **Loss Function**: Binary Crossentropy
- **Metrics**:
  - Accuracy
  - Precision
  - Recall

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6. Training Process

- Training performed using `model.fit()`
- Validation data used during training
- **EarlyStopping** applied:
  - Stops training when no improvement
  - Prevents overfitting

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7. Evaluation Metrics

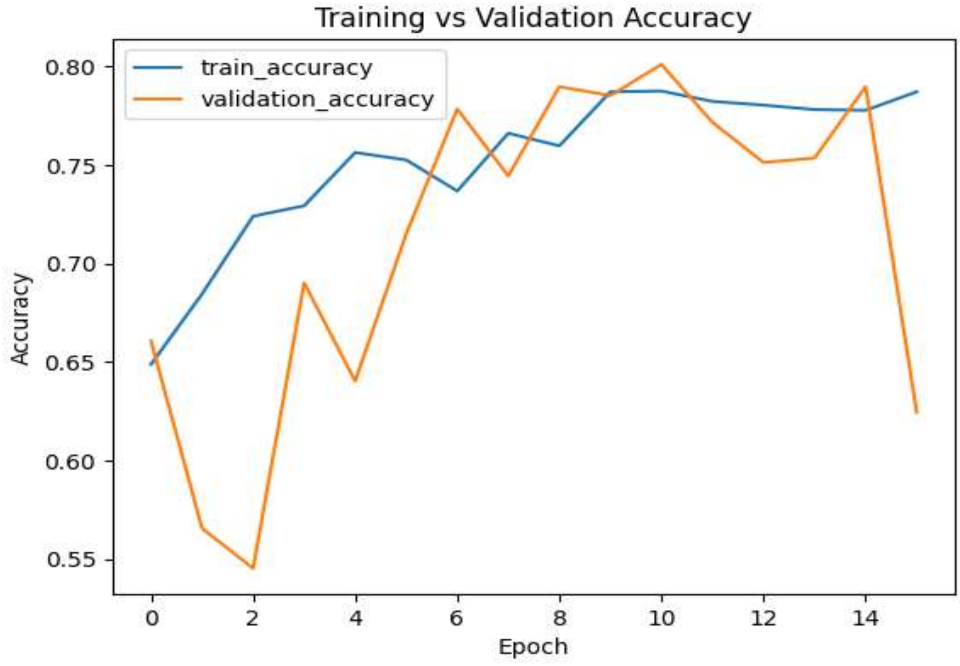
For Classification:

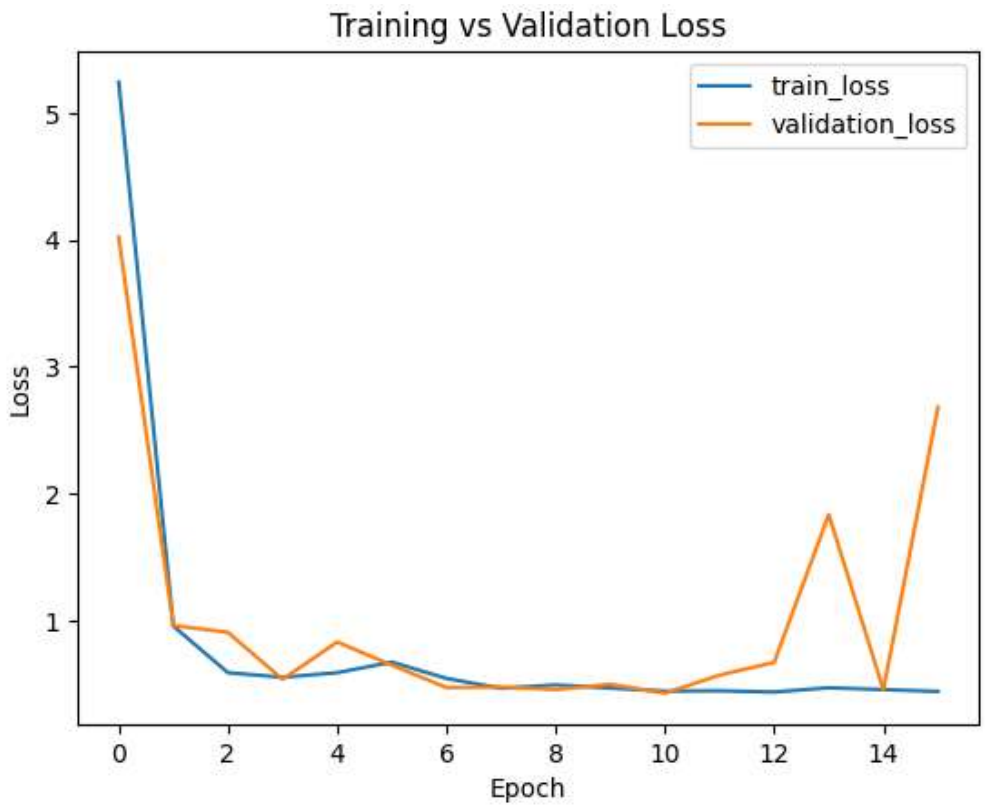
- Accuracy
- Precision
- Recall
- F1 Score

F1 Score:
F1 = 2 × (Precision × Recall) / (Precision + Recall)

CNN (Convolutional Neural Network)

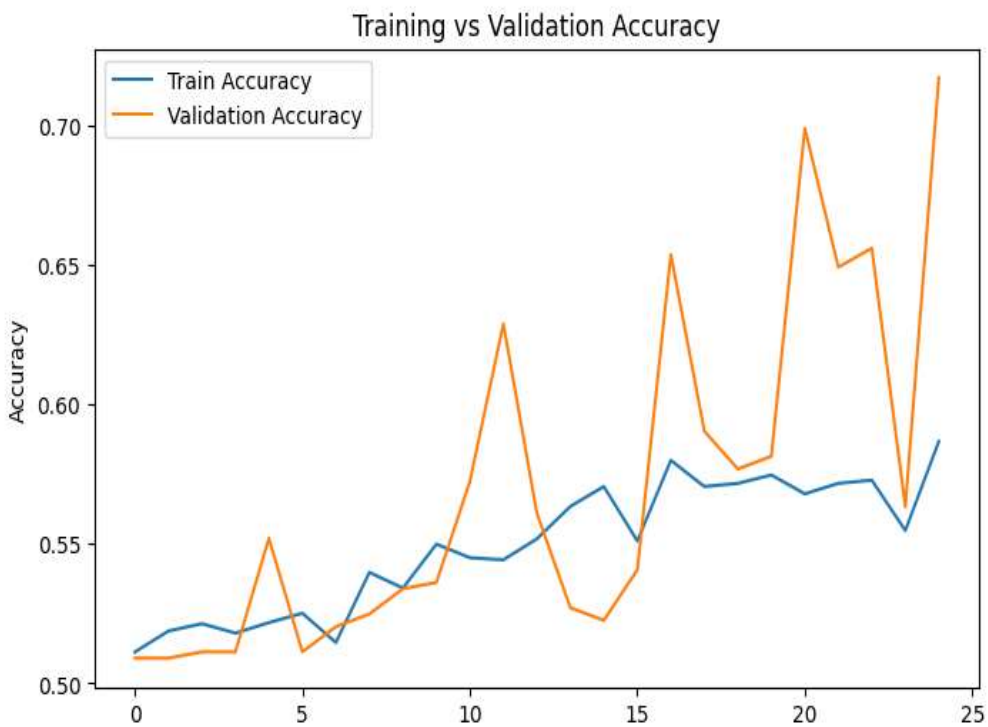
```
TRAIN RESULTS
------------------------------
Accuracy  : 0.8347
Precision : 0.9000
Recall    : 0.7284
F1 Score  : 0.8051
Loss      : 0.3559

VALIDATION RESULTS
------------------------------
Accuracy  : 0.8009
Precision : 0.8374
Recall    : 0.7556
F1 Score  : 0.7944
Loss      : 0.4279

TEST RESULTS
------------------------------
Accuracy  : 0.8279
Precision : 0.8701
Recall    : 0.7128
F1 Score  : 0.7836
Loss      : 0.3441
```

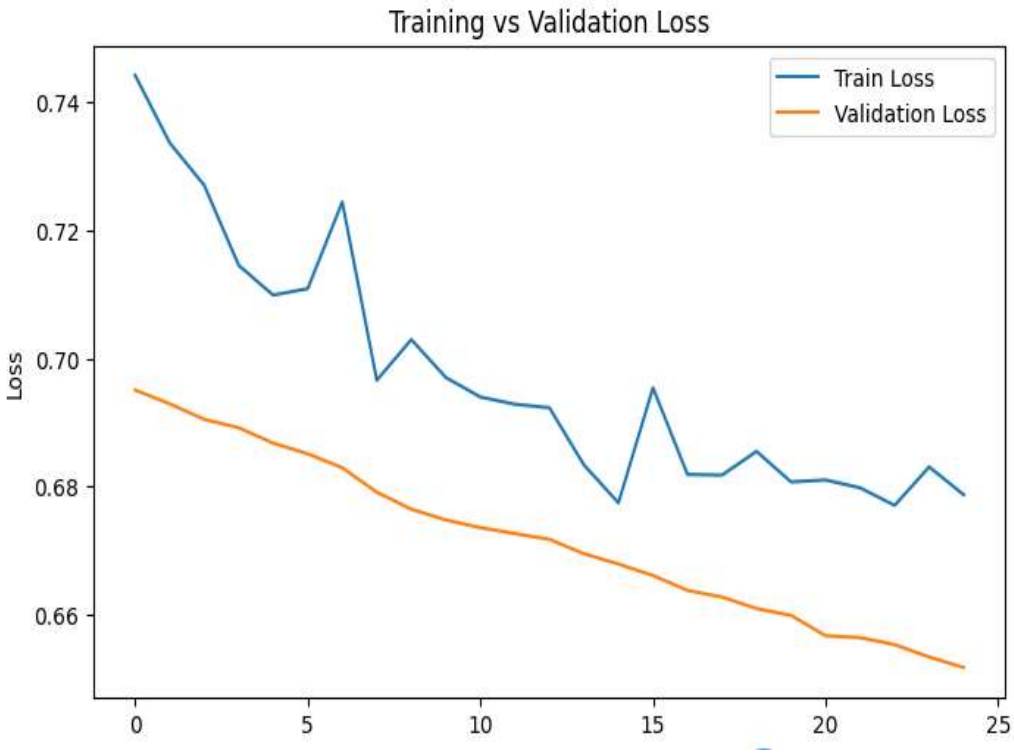

Training vs Validation Accuracy
train_accuracy
validation_accuracy
Accuracy
0.80
0.75
0.70
0.65
0.60
0.55
0
2
4
6
8
10
12
14
Epoch


Training vs Validation Loss
train_loss
validation_loss
Loss
5
4
3
2
1
0
2
4
6
8
10
12
14
Epoch

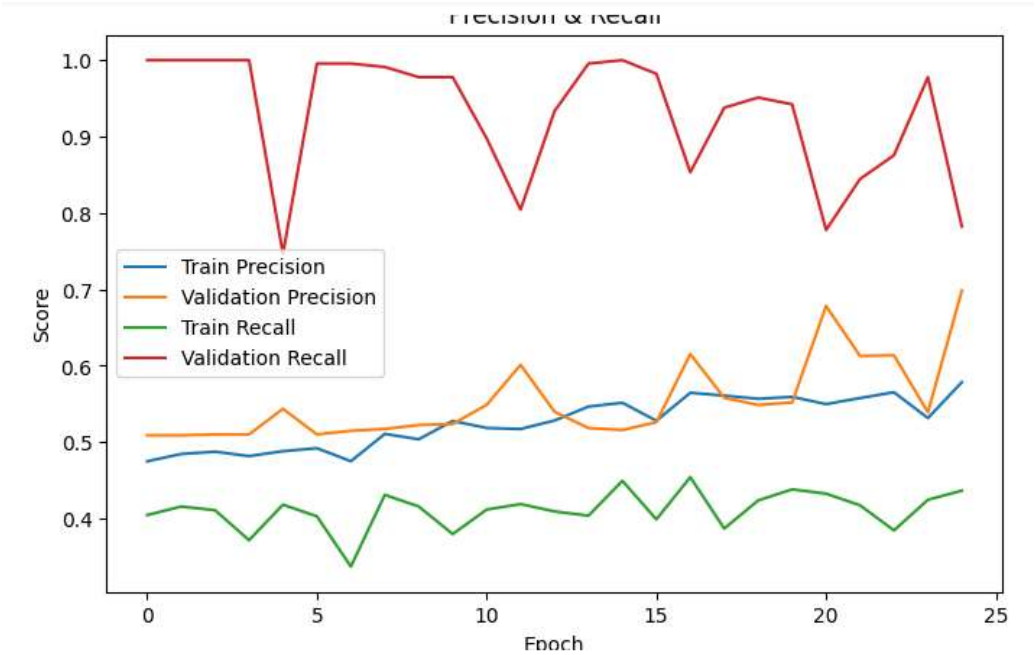
Transfer Learning Models

```
TRAIN RESULTS
------------------------------
Accuracy  : 0.6788
Precision : 0.6463
Recall    : 0.6955
F1 Score  : 0.6700
Loss      : 0.6662

VALIDATION RESULTS
------------------------------
Accuracy  : 0.7172
Precision : 0.6984
Recall    : 0.7822
F1 Score  : 0.7379
Loss      : 0.6517

TEST RESULTS
------------------------------
Accuracy  : 0.7070
Precision : 0.6449
Recall    : 0.7340
F1 Score  : 0.6866
Loss      : 0.6548
```





Precision & Recall Graph



For YOLO:

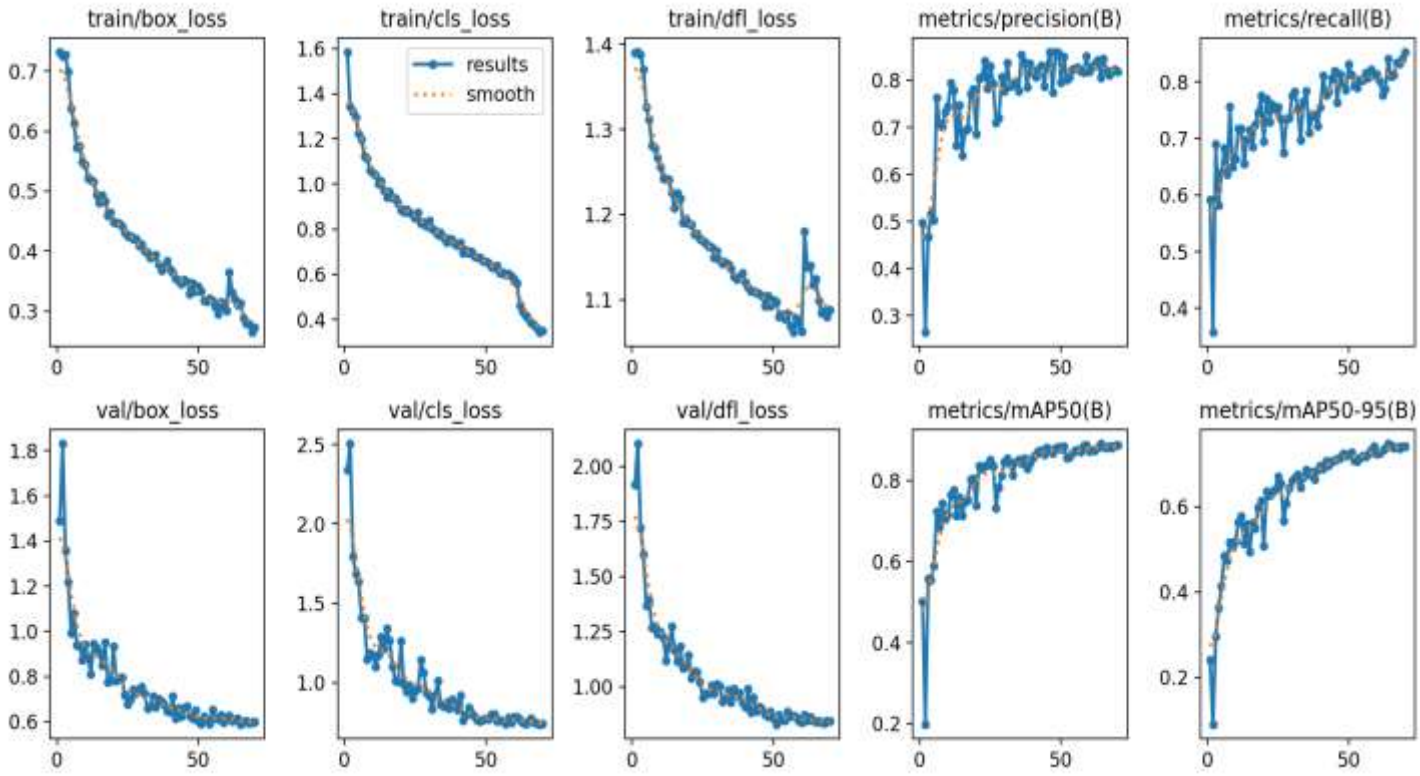
- mAP (mean Average Precision)
- Precision
- Recall

Training

```
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
                     all        442        442      0.806       0.84      0.891      0.747
                    bird        217        217      0.781      0.808      0.852       0.66
                   drone        225        225      0.831      0.873       0.93      0.834
  Speed: 2.0ms preprocess, 8.8ms inference, 0.0ms loss, 1.0ms postprocess per image
  Results saved to /content/runs/detect/val
  Precision: 0.8061433501168087
  Recall: 0.8402421681882101
  mAP50: 0.8908829900054251
  mAP50-95: 0.746787192330659
```

```
VALIDATION RESULTS
----------------------------
Accuracy  : 0.8232
Precision : 0.8061
Recall    : 0.8402
F1 Score  : 0.8228
```

```
TEST RESULTS
----------------------------
Accuracy  : 0.8310
Precision : 0.8362
Recall    : 0.8258
F1 Score  : 0.8310
```



YOLO training results graph

Overall Understanding

Model is:

Learning correctly
Improving over time
No major overfitting

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1. TRAINING LOSS GRAPHS (Top Left)

`train/box_loss`

- Decreasing from ~0.75 → ~0.27
  Means:
  Bounding box prediction is improving
  Model is learning object location

`train/cls_loss`

- Decreasing from ~1.6 → ~0.35
  Means:
  Classification (bird vs drone) is improving

`train/dfl_loss`

- Smooth decrease
  Means:
  Better bounding box refinement

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2. VALIDATION LOSS (Bottom Left)

`val/box_loss`, `val/cls_loss`, `val/dfl_loss`

- All decreasing gradually

Very important:
Validation loss follows training loss
No big gap → **No overfitting**

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3. PRECISION & RECALL (Top Right)

Precision

- Increasing (~0.5 → ~0.83)

Meaning:
Fewer false positives
Model predicts correctly

Recall

- Increasing (~0.6 → ~0.85)

Meaning:
Detects most objects
Fewer misses

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4. mAP GRAPHS (Bottom Right)

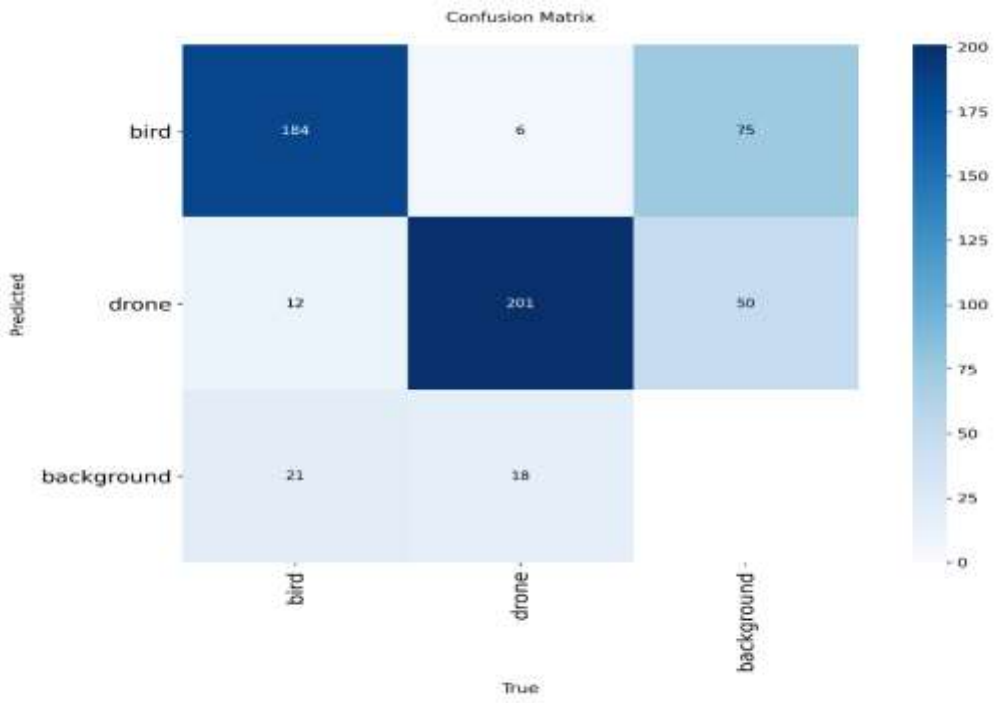
mAP@50

- ~0.89 (very good)

mAP@50-95

- ~0.75

Meaning:
Strong detection performance
Good bounding box accuracy



IMPORTANT INTERPRETATION

Model shows:

Stable learning
Good generalization
Balanced precision & recall
High detection accuracy

Small Observation

Slight fluctuations in early epochs:

- Normal (model learning phase)
- Not a problem

Final Conclusion (USE THIS IN REPORT)

The training and validation loss curves show a consistent decreasing trend, indicating effective learning. Precision and recall values increase steadily, demonstrating improved detection capability. The mAP@50 and

mAP@50-95 scores confirm that the model achieves strong performance in both classification and localization tasks. No significant overfitting is observed, as validation metrics closely follow training metrics.

Simple One-Line Answer

“Loss decreases and mAP increases, which shows the model is learning well without overfitting.”

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8. Results & Analysis

Accuracy Analysis

- Training accuracy increases steadily
- Validation accuracy slightly lower

Indicates good learning with mild overfitting

Loss Analysis

- Training loss decreases
- Validation loss stabilizes

Model is learning meaningful patterns

YOLO Performance

- Successfully detects objects in images
- Good bounding box predictions
- Real-time detection capability

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9. Observations

- Transfer Learning outperforms basic CNN
- EfficientNetB0 provides best balance of accuracy and efficiency
- YOLO is effective for object detection tasks
- EarlyStopping improves generalization

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10. Conclusion

- CNN is useful for basic image classification
- Transfer Learning significantly boosts performance
- YOLO enables real-time object detection
- Proper dataset splitting is crucial

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11. Future Work

- Fine-tuning pre-trained models
- Deployment using web

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12. Final Summary

This project demonstrates:

- Image classification using CNN & Transfer Learning
- Object detection using YOLO
- Performance evaluation using multiple metrics

The combination of these techniques provides a powerful solution for real-world computer vision applications.